

Strong-Field Gravity in the Mesh Framework

From a Derived Weak-Field Metric to a Derived Newton Constant, Black-Hole Thermodynamics, and
Frame Dragging

A Technical Report of the Mesh Framework Programme

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Programme credit: the core Rope Hypothesis is due to Bill Gaede; this corpus develops and formalises it. · August 2026

Status: the results below are conditional and labelled Modeled — an active campaign, not a closed theory. What is new is that gravity's strong-field sector now produces derived (if conditional) numbers where it previously produced only matched forms, including the framework's first derived dependence of Newton's constant on its own primitives.

Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

This report documents the strong-field gravity campaign of the Mesh Framework — the body of work that carries the sector past its derived weak-field metric into black-hole interiors, horizon thermodynamics, frame dragging, and the origin of Newton's constant. Three results stand out. First, the sector's state is now settled: the weak-field metric is *derived* (not merely matched to Schwarzschild), the horizon has a mechanical mechanism, and the strong-field expansion is certified. Second, and for the first time in the corpus's history, **Newton's constant G has a derived dependence on the framework's own primitives** — conditional and labelled, but derived rather than assumed. Third, a black-hole thermodynamics built from the medium's reconnection dynamics yields a temperature, an entropy, and a "quantum of the ring" whose action is independent of the black hole's mass — and that action is written in the same form, $\text{tension} \times \text{area} \div c$, as the corpus's expression for $\hbar$ in the electromagnetic sector. Every result here is conditional (Modeled); the campaign is ongoing, and this report states the successes and the mechanisms rather than the full chain of superseded attempts behind them.

1. The State of the Sector

For much of the programme's life the honest summary of gravity was modest: the framework could match the weak field but not derive it, and strong-field questions were out of reach. That summary is now stale, and the correction is itself a registered result. Three facts define where the sector actually stands:

The weak-field metric is derived. A counting argument closes it: the wave operator carries four coefficient functions, a static metric carries four, and the map between them is an exact bijection — leaving no room for a second, competing metric. Together with the classical weak-field tests (light deflection at 1.751 arcseconds, the perihelion precession, the redshift), the metric is a derived consequence of the medium's mechanics, not a form fitted to Schwarzschild.

The horizon has a mechanism. A black-hole horizon is the *tension-exhaustion surface* — the surface where the medium's capacity to carry a transverse wave reaches zero. This is a physical mechanism for the horizon, developed in the companion black-holes report, and it is the foundation the thermodynamics of Section 3 is built on.

The strong-field expansion is certified. The extrapolation that carries the sector's results from weak into strong field passed its audit — the gate opened — so the campaign's strong-field claims rest on a checked expansion rather than an unverified leap.

2. Newton's Constant, Derived From the Medium

The headline result of the campaign is that Newton's constant G, for four hundred claims an assumed input, now has a **derived dependence on the framework's own primitives**. The derivation multiplies three registered results: the induced-stiffness coefficient measured from the medium's own vibrational

band, the speed-of-light anchor that fixes the band's absolute frequency scale, and a locking condition that leaves the band's shape no free modulus to depend on. The product is

$$G = c^3 a^2 / (16\pi \zeta \hbar), \text{ with } \zeta = 1.208$$

where a is the mesh scale. The result is *tension-inert and spacing-quadratic*: G does not depend on the medium's tension, and depends on the mesh spacing as a^2 . This is the corpus's first derived exponent pair for G — $(0, 2)$ — and it carries its three premises in its name rather than hiding them. It is the mechanism by which gravity's weakness is expressed in the framework: G is not small by fiat but small because it is set by the mesh scale squared against c^3 .

The honest label matters. This is a conditional derivation — conditional on the Sakharov (induced-gravity) reading under which the medium's vibrational vacuum generates the gravitational action, and on the band-scale inputs it inherits. It is not a parameter-free prediction of G 's numerical value. What changed is the *kind* of statement the sector can make: from “ G is an input” to “ G 's dependence on the primitives is derived, given these named premises.” That is a real move, and the campaign made it deliberately, with the bars for the derivation locked before the computation ran.

3. Black Holes as Reconnection Engines

Beyond the horizon mechanism (Section 1), the campaign built a *thermodynamics* for the black-hole interior out of the medium's reconnection dynamics — the discrete crossing-and-rejoining events the framework already uses for entropy counting. The chain runs from a measured bit-cost to a temperature to a striking mass-independent constant.

A temperature with a mechanism. The interior behaves as a two-state reservoir whose temperature is the oldest formula in thermodynamics — an energy gap over the log-odds of occupation — verified by machine ($dE = T \cdot dS$ to numerical precision). The energy cost of writing one reconnection bit tracks the crossing threshold exactly (measured at $15.67 \times$ the threshold energy, stable to three digits across a factor-of-four sweep), which is what gives the temperature its value a mechanism rather than a fit.

A quantum of the ring, independent of the black hole's mass. The campaign's signature move is a series of cancellations in which the black-hole's specific parameters drop out of the physics one by one. The last of these is the sharpest: the action of a single reconnection “snap” at the horizon is independent of both the local depth and the black hole's mass — the mass and the surface-gravity scale cancel exactly, by machine. The horizon therefore carries a fixed quantum of action per event, the same for a stellar black hole and a supermassive one. A mass-independent action emerging from a mass-dependent object is exactly the kind of universal constant a fundamental theory should produce.

The same hand as the fine-structure constant. The most suggestive result of the campaign is a coincidence written down *before* it was evaluated, as the discipline required. The corpus's expression for $\hbar$ in the electromagnetic sector and the horizon's emergent snap action are the *same form* — both are tension $\times$ area $\div c$. Gravity's horizon and electromagnetism's fine structure, two sectors that were never built to consult each other, write their actions in the same hand. Divide the two and the dimensional scaffolding falls away, leaving the fine-structure constant times a ratio of two areas. Whether this is a deep unity or a structural coincidence is not yet settled; that it was pre-registered and survived evaluation is why it is in the record.

4. Frame Dragging and the Twist Sector

A rotating mass should drag the local inertial frame around with it — the Lense-Thirring effect, measured by Gravity Probe B and by satellite laser ranging. The campaign’s route to frame dragging runs through the medium’s own twist sector: the rotational degree of freedom the two-strand medium carries natively. The rotating source sets up a stationary twist configuration with the correct radial and angular structure for a frame-dragging field.

The result the arc rested on was already in the corpus. A derived theorem — that a static defect sources zero net torque and zero net winding force at dipole order — is exactly the structural constraint the frame-dragging field must satisfy, and it puts two of the arc’s requirements on derived footing. The confrontation with the Lense-Thirring data is not yet decisive: the framework’s quantitative position depends on coefficients still being resolved, and the report states that openly. What is established is the *mechanism* — frame dragging as the medium’s twist sector responding to a rotating source — and its grounding in a derived theorem rather than an assumed form.

5. Honest Status, and What Remains Open

Every result in this report is labelled Modeled: conditional, with its premises named, and part of an active campaign rather than a closed sector. That labelling is deliberate and it is the point of the framework’s discipline. Specifically:

Result	What is established	What is conditional / open
Weak-field metric	Derived by the bijection counting argument; classical tests unconditional.	— (this half is settled)
Derived G	G’s dependence on the primitives is derived: $G = c^3a^2/(16\pi\zeta\hbar)$, exponent pair (0,2).	Conditional on the Sakharov induced-gravity reading and the band-scale inputs; not a numerical prediction of G.
BH thermodynamics	Two-state temperature verified by machine; bit-cost measured; mass-independent snap action.	Coefficients still being pinned; the $\text{tension}\times\text{area}\div c$ coincidence with $\hbar$ is suggestive, not yet explained.
Frame dragging	Mechanism (twist sector) grounded in a derived zero-torque theorem.	Quantitative Lense-Thirring confrontation not yet decisive; coefficients open.

The through-line of the campaign is a single shift in what the sector can say. Where gravity once offered matched forms, it now offers derived numbers — conditional ones, with their premises on their face, but derived. The first derived G, a black-hole thermodynamics with a mass-independent action, and a frame-dragging mechanism grounded in a registered theorem are the concrete evidence of that shift. The work is not finished, and this report does not pretend it is; it documents what has been established so that the progress is visible outside the claims registry and the code that produced it.

Provenance: the results here are registered as claims GRV-054 through GRV-095 in claims.yaml, with per-session bars and benchmarks in the analysis/ and benchmarks/gravity/ directories. This report is a companion to the weak-field gravity paper and the black-holes paper, covering the strong-field material that postdates them.